SQL SNACKS

Lagos Ride-Sharing Analytics Challenge

semicolon

Database Schema

Key Tables:

1. **Drivers**:
   - `DriverID`, `Name`, `Rating`, `TotalRides`
2. **Riders**:
   - `RiderID`, `Name`, `City`
3. **Rides**:
   - `RideID`, `DriverID`, `RiderID`, `Fare`, `Rating`, `DistanceKM`
4. **Payments**:
   - `PaymentID`, `RideID`, `Amount`, `PaymentMethod`

Insert Data

Data to be inserted into the tables are in the document below:
https://docs.google.com/document/d/1cWiodbGyT7iZ_YydY55DZ0ttWtxxZEM0BxX4yIQ13aY/edit?usp=sharing

The data was gathered over the course of one month

Basic Tasks

NOTE: Save the results in a table as the tables will be used later when visualising the data.

1. **Top 5 highest-rated drivers in Lagos**
2. **Riders with more than 5 rides in the last 30 days in Lagos**
3. **Total revenue for the past month for Lagos rides**
4. **Area in Lagos with the most rides**

Advanced Queries

5. **Driver with the highest revenue in Lagos**
   Expected result table:

DriverID	Name	Total Revenue

Advanced Queries

6. **Find rides where the fare is more than 50% higher or lower than the average fare**

Expected result table:

RideID	Fare	Average Fare	Average Fare

Advanced Queries

7. **Find riders in Lagos who rated their drivers less than 3 on average**

Expected result table:

RideID	Name	Average Rating

Additional Tasks

8. **Find the top 5 Lagos neighborhoods with the highest average fare**

Expected result table:

Area	Average Fare

Additional Tasks

9. **Drivers in Lagos who have not received any rides in the last 30 days**

Expected result table:

DriverID	Name

Additional Tasks

10. **Rides in Lagos with the longest distance (top 5)**

Expected result table:

RideID	DistanceKM	DriverID	RiderID

Optimization Queries

11. **Find the number of rides each driver in Lagos has had, sorted by the most rides**

Expected result table:

DriverID	Total Rides

Optimization Queries

12. **Identify the payment methods used by Lagos riders for rides over ₦50,000**

Expected result table:

Payment Method	Number of Transaction

Optimization Queries

13. **Find rides in Lagos with a duration longer than 2 hours**

Expected result table:

RideID	Duration	DriverID	RideID

For the ambitious ones

Data Visualization Ideas

1. Bar Chart: Top-rated Drivers in Lagos

- **Data Source:**
  - **Tables:** `Rides`, `Drivers`
  - **What to Extract:** Average ratings for each driver.

2. Pie Chart: Payment Methods for Rides Over ₦50,000

- **Data Source:**
  - **Tables:** `Rides`, `Payments`
  - **What to Extract:** Count of each payment method for rides with fares > ₦50,000.

Data Visualization Ideas

3. Line Graph: Monthly Revenue Trends in Lagos

- **Data Source:**
  - **Tables:** `Rides`, `Payments`
  - **What to Extract:** Monthly total revenue from rides.

4. Heatmap: Most Popular Lagos Areas Based on Rides

- **Data Source:**
  - **Table:** `Rides`
  - **What to Extract:** Count of rides in each popular area.

Deliverables

SQL Scripts:

- Create tables, populate data, and write queries.

Results Table:

- Store query results for analysis.

Visualizations (Optional):

- Create graphs/charts based on the query results.

NOTE: The link to the Excel sheet should be added to the “our links sheet”.

Submission on the 17th of April, 2025.

Enjoy!!!

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